EgoBlind-RA: Towards Safer Egocentric Assistive AI for Blind Users via Risk-Adaptive Routing

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Abstract

Blind and low-vision (BLV) users depend on visual assistance systems to navigate an unpredictable physical world, yet existing multimodal large language models (LLMs) apply a uniform inference policy regardless of query urgency. We present **EgoBlind-RA**, a risk-adaptive framework that classifies egocentric query-video pairs as *urgent* or *non-urgent* using a lightweight CLIP-based classifier, then routes each query to a response policy calibrated to its safety stakes via urgency-conditioned LoRA fine-tuning and Direct Preference Optimization (DPO). We evaluate EgoBlind-RA on the EgoBlind benchmark using a vision-language baseline (Kimi-VL) and urgency-conditioned LoRA fine-tuning via LLaMA-Factory. Our CLIP-based urgency classifier achieves 0.905 ROC-AUC on the held-out test set. Baseline evaluation across 4,029 examples reveals that model performance is weakest on safety warnings and navigation queries, precisely the categories where errors carry the greatest risk, validating the need for urgency-aware routing. Initial fine-tuning experiments surface a critical template-mismatch failure mode in adapter training, which we analyze and propose to resolve in future work.

1 Introduction

For the ∼253M people worldwide living with visual impairment, AI-powered assistive systems represent a transformative opportunity to support independence and safety in daily life [1]. The emergence of large language models (LLMs) capable of jointly processing vision and language has opened new possibilities for blind and low-vision (BLV) users, enabling systems that interpret first-person video and answer natural language queries about the user's immediate environment. Recent benchmarks including EgoBlind [2] and visual question answering (VQA) datasets such as VizWiz [3] have begun to formalize this setting, revealing unique challenges posed by egocentric BLV queries: off-center regions of interest, ambiguous spatial context, and information needs that range from casual to life-critical.

The central challenge we address is that existing multimodal systems treat all queries uniformly, regardless of safety stakes. A user asking *"Is there a step in front of me?"* faces an immediate physical hazard demanding a fast, conservative response; a user asking *"What brand of cereal is this?"* does not. State-of-the-art LLM-based systems such as LLaVA [4] and InstructBLIP [5] apply a single inference policy to both query types. Such a gap is consequential: excessive caution degrades usability for routine queries, while insufficient urgency in safety-critical responses can cause physical harm. Standard VQA benchmarks do not evaluate this dimension, leaving the problem unaddressed in the literature.

In this paper, we propose **EgoBlind-RA**, a risk-adaptive egocentric visual assistance framework for BLV users. We introduce a two-stage pipeline: a lightweight CLIP-based urgency classifier that labels each query-video pair as *urgent* or *non-urgent*, followed by a routing mechanism that

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selects a response policy matched to the predicted safety stakes. We investigate two architectures: (1) hard-routing to two specialized LoRA-fine-tuned adapters, and (2) soft-routing via urgency tags to a single unified adapter, thereby enabling a direct comparison of specialization versus generalization under a risk-aware objective. Evaluated under the EgoBlind benchmark, it is our hope that our approach improves response safety for urgent queries and answer quality for non-urgent queries relative to a uniform-policy baseline.

2 Related Work

Visual assistance for BLV users. VizWiz [3] established the BLV visual assistance domain by crowd-sourcing answers to photos taken by BLV users, exposing challenges unique to this population: poor image framing, motion blur, and high-stakes information needs. Gurari et al. [6] extended this with a large-scale VQA benchmark, demonstrating that BLV-generated images are significantly harder for models to understand than standard VQA datasets. More recently, EgoBlind [2] introduced the first egocentric video benchmark for BLV assistance. Our work builds on EgoBlind but introduces urgency-conditioned response policies absent from prior work.

Egocentric video understanding. Ego4D [7] catalyzed first-person perception research with a large-scale egocentric corpus spanning episodic memory and forecasting. Embodied QA [8] further explored grounded language understanding in interactive environments. These works, however, target able-bodied users and do not model the safety-critical nature of BLV queries.

Multimodal LLMs. LLaVA [4] and InstructBLIP [5] demonstrated strong zero-shot performance on visual QA tasks by aligning vision encoders with LLMs. Parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) [9] has made domain adaptation of such models tractable at modest compute. We apply LoRA fine-tuning in both single-adapter and dual-adapter configurations for the BLV assistance domain.

Alignment and preference optimization. Reinforcement learning from human feedback (RLHF) [10] is a standard technique for aligning language model outputs with human preferences. Direct Preference Optimization (DPO) [11] offers a stable, reward-model-free alternative, which we extend to an urgency-conditioned setting by constructing preference pairs that reflect both response quality and latency constraints, weighted by urgency regime.

Risk-aware inference. Selective abstention [12], semantic entropy [13], and conditional computation [14] share our motivation of matching model behavior to input risk, but none targets safety-critical assistive settings with explicit urgency signals. To our knowledge, EgoBlind-RA is the first system to jointly address urgency classification, risk-stratified routing, and urgency-conditioned alignment for egocentric visual assistance.

3 Problem Statement

Let $\mathbf{x} = (v, q)$ denote a video-query pair, where $v \in \mathcal{V}$ is an egocentric video clip and $q \in \mathcal{Q}$ is a natural language query from a BLV user. Each pair is associated with a binary urgency label $u \in \{0, 1\}$ and a reference answer set $\mathcal{Y}^* = \{y_1^*, \ldots, y_K^*\}$. EgoBlind-RA solves two coupled subproblems:

(1) learn an **urgency classifier** $f_\theta : \mathcal{V} \times \mathcal{Q} \to \{0, 1\}$, ideally penalizing false negatives more heavily given the safety cost of missed urgent queries;

(2) learn a **conditional response generator** $g_\phi : \mathcal{V} \times \mathcal{Q} \times \{0, 1\} \to \mathcal{A}$, whose objective depends on the predicted urgency $\hat{u} = f_\theta(\mathbf{x})$.

In particular, for *urgent* queries ($\hat{u} = 1$), the response $y = g_\phi(\mathbf{x}, 1)$ is optimized under the composite loss:

$$\mathcal{L}_{\text{urgent}} \;=\; \alpha\,\mathcal{L}_{\text{acc}} + \beta\,\mathcal{L}_{\text{util}} + \gamma\,\mathcal{L}_{\text{lat}}, \qquad \alpha, \beta, \gamma > 0,\; \alpha + \beta + \gamma = 1, \tag{1}$$

where $\mathcal{L}_{\text{acc}}$ penalizes semantic mismatch against the reference set $\mathcal{Y}^*$, $\mathcal{L}_{\text{util}}$ penalizes low assistive utility as well and $\mathcal{L}_{\text{lat}}$ penalizes excessive token count as a differentiable proxy for generation latency.

By contrast, for *non-urgent* queries ($\hat{u} = 0$), $\mathcal{L}_{\text{lat}}$ is dropped and the model optimizes $\mathcal{L}_{\text{acc}}$ and $\mathcal{L}_{\text{util}}$ alone. We discuss each loss component in detail in Appendix A.

4 Proposed Approach

EgoBlind-RA operates as a two-stage pipeline: (i) an urgency classifier predicts the risk level of each query, and (ii) a conditional response generator produces an answer whose objective is matched to the predicted urgency regime. We investigate two instantiations of the second stage.

4.1 Stage 1: Urgency Classification

Given a video-query pair $\mathbf{x} = (v, q)$, a lightweight classifier f_θ predicts the binary urgency label $\hat{u} \in \{0, 1\}$. We implement f_θ as a frozen CLIP ViT-B/32 backbone whose frame and text embeddings are concatenated and passed through a two-layer MLP head, trained with standard binary cross-entropy loss (Section 6.1). The classifier currently uses a default decision threshold of 0.5; as discussed in Section 7, threshold tuning and asymmetric loss weighting are planned improvements to reduce the false-negative rate on urgent queries. The full classifier implementation is available in the project repository.[2]

4.2 Stage 2: Urgency-Conditioned Response Generation

Both approaches below share the same composite loss defined in Section 3. The key design variable is *how* urgency conditions the generator.

Approach 1: Modular hard-routing (bifurcated adapters). We train two separate LoRA adapters on a shared base model $\mathcal{M}$. The *urgent adapter* ϕ_u is fine-tuned on the subset $\mathcal{D}_u = \{(\mathbf{x}_i, y_i) : u_i = 1\}$ using the full loss $\mathcal{L}_{\text{urgent}}$, which includes the latency penalty $\mathcal{L}_{\text{lat}}$ to encourage concise, actionable responses. Training targets are the *shortest* correct reference answer for each example, biasing the model toward brevity. The *non-urgent adapter* ϕ_n is fine-tuned on $\mathcal{D}_n = \{(\mathbf{x}_i, y_i) : u_i = 0\}$ using only $\mathcal{L}_{\text{acc}} + \mathcal{L}_{\text{util}}$, with the *longest* reference answer as target to encourage thorough, detailed responses. At inference time, the classifier prediction $\hat{u}$ determines which adapter is loaded:

$$y = \begin{cases} \mathcal{M}_{\phi_u}(\mathbf{x}) & \text{if } \hat{u} = 1, \\ \mathcal{M}_{\phi_n}(\mathbf{x}) & \text{if } \hat{u} = 0. \end{cases} \tag{2}$$

This hard-routing design allows each adapter to specialize its generation behavior without compromising the other regime.

Approach 2: Prompt-conditioned unified adapter. A single LoRA adapter ϕ_{uni} is trained on the full dataset $\mathcal{D}_u \cup \mathcal{D}_n$, with the urgency label injected as a control tag prepended to the system prompt:

$$\texttt{[URGENT]} \text{ — optimize under } \mathcal{L}_{\text{urgent}}, \tag{3}$$

$$\texttt{[NON-URGENT]} \text{ — optimize under } \mathcal{L}_{\text{acc}} + \mathcal{L}_{\text{util}}. \tag{4}$$

The loss function switches dynamically per example based on the prepended tag. This tests whether a single model can internalize risk-adaptive behavior through prompt conditioning alone, without architectural specialization.

4.3 Base Model and Fine-Tuning

We select **Kimi-VL-A3B-Instruct** [15] as the base model: a 17.5B-parameter mixture-of-experts vision-language model with 2.8B active parameters, balancing capacity with the compute constraints of academic hardware. Fine-tuning is performed via LoRA [9] using LLaMA-Factory [16], with rank $r = 16$, $\alpha = 32$, applied to all attention and feed-forward projections (`q,k,v,o,gate,up,down_proj`), yielding ∼287M trainable parameters (<2% of total). Training uses AdamW with a cosine learning rate schedule (lr $= 2 \times 10^{-5}$), batch size 1 with gradient accumulation over 16 steps, for 5 epochs on a single NVIDIA L40S GPU (48 GB).

[2] `https://github.com/juliavekim/EgoBlind-RA`

Text-only SFT. As an initial experiment toward the full vision-conditioned pipeline, we perform supervised fine-tuning (SFT) using *text-only* inputs (question $\rightarrow$ answer), omitting video frames. This isolates the effect of urgency-conditioned objectives on language generation behavior before introducing visual grounding, and is computationally cheaper to iterate on.

4.4 Evaluation via Composite Loss

All generated responses are scored using the composite loss from Section 3, with similarity S implemented as token-overlap F1 (a lightweight proxy for semantic similarity). We conduct an extensive grid search over hyperparameters $(\alpha, \beta, \gamma, \tau_{\min}, \tau_{\max}, p, \kappa)$ across 3,888 configurations to identify the loss parameterization that best separates urgent from non-urgent performance, selecting the configuration that minimizes overall test loss. Full loss component definitions appear in Appendix A.

5 Experimental Methodology

Dataset. We use the EgoBlind dataset [2], comprising egocentric video clips paired with natural language questions from BLV users, with up to four reference answers per question and six question categories: information reading, safety warnings, navigation, social communication, tool use, and other resources. Since EgoBlind does not include urgency metadata, we annotate all examples using GPT-5.2 (`gpt-5.2-2025-12-11`, `reasoning_effort=xhigh`), providing 5 video frames per clip and prompting the model to classify each query as *urgent* or *not_urgent* from the perspective of a blind user. Using the question and video context, we manually verified a subset of 300 questions, finding >90% agreement.

Data splits. The training set contains 2,746 questions across 922 videos (1,474 urgent, 1,272 non-urgent) and the test set contains 1,283 questions (668 urgent, 615 non-urgent). For the CLIP classifier, we further split the training set into train/validation partitions via stratified sampling.

Baseline. We evaluate a uniform-policy baseline using `moonshot-v1-32k-vision-preview` (Kimi API). For each example, 5 frames leading up to the posed question are extracted on-the-fly from the source video (resized to 320px width, JPEG quality 60) and submitted alongside the question text. Urgency-specific system prompts instruct the model to produce concise responses for urgent queries and detailed responses for non-urgent queries, but no fine-tuning or routing is applied; the same model processes all queries. All 4,029 examples are evaluated ($\sim$21.4M tokens, $\sim$ \$25).

Fine-tuning (Approach 1). For the bifurcated adapter experiment, we train two LoRA adapters on the text-only subset as described in Section 4. Urgent SFT trains on 1,474 examples for 465 steps (3h 46m); non-urgent SFT trains on 1,272 examples for 400 steps (3h 05m). Training is conducted on MIT's Engaging cluster using SLURM-allocated L40S GPUs.

Evaluation metrics. We report token-overlap F1 as a proxy for semantic similarity S, the composite loss $\mathcal{L}$ under the best grid-search configuration ($\alpha = 0.5, \beta = 0.3, \gamma = 0.2, \tau_{\min} = 8, \tau_{\max} = 40, p = 3, \kappa = 0.3$, subject to $\alpha + \beta + \gamma = 1$), and per-category breakdowns. For the urgency classifier, we report accuracy, precision, recall, F1, and ROC-AUC.

6 Results and discussion

6.1 CLIP Urgency Classifier

Architecture and training. We implement a CLIP-based binary urgency classifier (`ViT-B/32`, pretrained on OpenAI weights) that encodes both the video and associated natural language query. For each query timestamp t_q, $K = 4$ frames are uniformly sampled from ± 2-second temporal window centered at t_q. Frame embeddings are mean-pooled and concatenated with the CLIP text embedding of the question, then passed through a lightweight two-layer MLP classifier. The CLIP backbone is frozen throughout training; only the MLP head is updated. The CLIP backbone is frozen throughout; only the MLP head is updated via AdamW (lr $= 10^{-4}$) with binary cross-entropy loss over 5 epochs on an NVIDIA L40S GPU.

Results. Table 1 in Appendix B reports classifier performance on the held-out validation and test sets. The model learns a strong urgency signal despite its frozen backbone, achieving a ROC-AUC of 0.938 on validation and 0.905 on test, indicating reliable discriminative ability across splits. The primary failure mode is recall on the urgent class (0.695 on test), reflecting a tendency to *under-predict* urgency that is especially costly in safety-critical deployment.

Discussion. The strong ROC-AUC across both splits suggests the frozen CLIP backbone captures a meaningful urgency signal from the joint video-text representation, even without end-to-end fine-tuning. The modest accuracy gap between validation (86.3%) and test (79.8%) is expected given that the test set contains entirely unseen videos. The main limitation is that the model misses $\sim$ 30% of urgent queries on the test set, mislabeling them as non-urgent, which is the most costly error in a safety-critical assistive setting. Future work could explore threshold tuning, asymmetric loss weighting, or partial backbone unfreezing to improve recall without sacrifice to precision.

6.2 Baseline Evaluation

Table 2 in Appendix B summarizes the uniform-policy baseline performance on the test set. Overall token-overlap F1 is 0.1025, with urgent queries (0.1245) outperforming non-urgent queries (0.0778), likely because urgent answers tend to be shorter and more lexically constrained, favoring our token-overlap metric.

Per-category analysis. Table 3 in Appendix B breaks down the best-configuration composite loss by question type. Safety warnings exhibit the highest loss (1.118), followed by tool use (0.948) and navigation (0.787). Information reading and social communication, which are predominantly non-urgent, have the lowest losses. This confirms the main finding of EgoBlind [2]: models struggle most on exactly the categories where errors are most dangerous, validating the need for urgency-aware routing.

Grid search. We evaluate all 4,029 baseline predictions across 3,888 hyperparameter configurations of the composite loss $(\alpha, \beta, \gamma, \tau_{\min}, \tau_{\max}, p, \kappa)$, of which 486 satisfy the normalization constraint $\alpha + \beta + \gamma = 1$. Among constrained configurations, the best by overall test loss is $\alpha = 0.5, \beta = 0.3, \gamma = 0.2, \tau_{\min} = 8, \tau_{\max} = 40, p = 3, \kappa = 0.3$ (loss $= 1.578$). The top 5 configurations all share $\alpha = 0.5, \beta = 0.3, \gamma = 0.2$ and $\tau_{\max} = 40$, varying only in $\tau_{\min}$ and p, suggesting the loss is relatively insensitive to latency-regime boundaries but sensitive to the accuracy–latency tradeoff weight. Notably, the optimal $\gamma = 0.2$ assigns meaningful weight to latency, confirming that brevity matters for urgent responses and that dropping the latency term for non-urgent queries (as our framework does) is a principled design choice.

6.3 Fine-Tuning Results and Failure Analysis

Training dynamics. Both LoRA adapters show healthy training loss curves: the urgent adapter converges from 1.83 to 0.48 over 465 steps, and the non-urgent adapter from 1.61 to 1.22 over 400 steps, indicating that the models successfully learn from the training signal.

Evaluation. Despite promising training dynamics, the fine-tuned models produce degenerate outputs at inference time. Table 4 in Appendix B compares the fine-tuned system against the baseline. Overall loss *increases* by 1.32, and urgent loss nearly doubles. Qualitative inspection reveals repeated tokens, language-switching artifacts (English $\rightarrow$ Chinese), and degenerate loops (e.g., "`I don't know. I don't know. I don't know.`").

Root cause: template mismatch. We trace the failure to a missing `template` parameter in the LLaMA-Factory training configuration. LLaMA-Factory uses model-specific chat templates to format training examples with the correct special tokens and role markers. Our configuration files omitted this parameter, causing LLaMA-Factory to apply a default template that does not match Kimi-VL's native chat format. Consequently, the adapter weights were optimized against incorrectly tokenized inputs. At inference time, when we applied Kimi-VL's native `apply_chat_template()`, the model encountered a distribution shift between training and inference formatting, producing degenerate outputs. Critically, this failure is *not* a modeling limitation—training loss decreased normally—but an engineering issue in the fine-tuning pipeline that we plan to resolve.

7 Next Steps and New Research Ideas

Our midterm results establish a solid experimental foundation: a validated urgency classifier, a comprehensive baseline with per-category analysis, and a fully implemented fine-tuning pipeline, while surfacing a critical template-mismatch bug that we now know how to fix. We outline the following next steps in priority order.

1. Fix the template mismatch and re-run SFT. The immediate priority is to register or identify the correct chat template for Kimi-VL in LLaMA-Factory's template registry, re-run both urgent and non-urgent SFT ($\sim$4 hours each on L40S), and re-evaluate. We expect this alone to produce coherent outputs and a meaningful comparison against the baseline, since training loss curves confirm the model learned from the data.

2. Direct Preference Optimization (DPO). Once SFT produces coherent outputs, we plan to apply DPO [11] to the urgent adapter. The pipeline is already implemented: for each urgent training example, we generate $k = 8$ candidate responses from the SFT model, score each with the composite loss $\mathcal{L}_{\text{urgent}}$, and select the lowest-loss response as "chosen" and the highest-loss as "rejected." DPO then aligns the model toward responses that jointly optimize accuracy, utility, and latency without requiring a separate reward model. This is a natural extension of SFT that directly incorporates our urgency-aware loss into the alignment objective.

3. Vision-conditioned fine-tuning. Our current SFT experiments use text-only inputs (question $\rightarrow$ answer) to isolate language-generation behavior. The next step is to incorporate video frames during fine-tuning, matching the inference-time input modality. This requires frame extraction and formatting within LLaMA-Factory's multimodal data pipeline, which we have already prototyped for the baseline evaluation.

4. Improved evaluation metrics. Token-overlap F1 is a coarse proxy for semantic similarity that penalizes valid paraphrases. We plan to upgrade to BERTScore F1 or an LLM-based judge (following EgoBlind's evaluation protocol) for the final evaluation. This is especially important for non-urgent responses, which tend to be longer and more varied in phrasing. However, our ability to to upgrade to an LLM-based judge is dependent upon us getting more compute credits from the department.

5. Approach 2 integration and comparison. The prompt-conditioned unified adapter provides the key experimental contrast with the bifurcated architecture. Once both approaches produce coherent fine-tuned outputs, we will compare them under identical evaluation conditions to test whether architectural specialization (Approach 1) outperforms prompt-based conditioning (Approach 2) on the urgency-aware composite loss.

6. Classifier improvements. The CLIP urgency classifier currently misses $\sim$30% of urgent queries on the test set (Table 1), using standard binary cross-entropy with no class weighting. Since false negatives are far more costly than false positives in this setting, we plan to (a) tune the decision threshold to maximize recall at a controlled precision level, (b) introduce asymmetric loss weighting to penalize missed urgent queries more heavily during training, and (c) explore partial backbone unfreezing to further reduce the false-negative rate.

7. End-to-end system evaluation. The final evaluation will measure the complete pipeline: classifier $\rightarrow$ router $\rightarrow$ regime-specific generator, reporting per-category composite loss, latency, and qualitative safety analysis. We will also analyze cascading errors where misclassification by the urgency classifier leads to misrouted responses, quantifying the downstream impact on safety-critical queries.

8 Code and Reproducibility

The implementation of **EgoBlind-RA** is publicly available on GitHub at github.com/juliavekim/EgoBlind-RA. Individual course repositories, containing course assignments and the EgoBlind-RA final project, are also available: Julia's and Xander's.

References

[1] Rupert R.A. Bourne et al. Magnitude, temporal trends, and projections of the global prevalence of blindness and distance and near vision impairment: a systematic review and meta-analysis. *The Lancet Global Health*, 5(9):e888–e897, 2017.

[2] Junbin Xiao et al. Egoblind: Towards egocentric visual assistance for the blind. *arXiv preprint arXiv:2503.08221*, 2025.

[3] Jeffrey P. Bigham et al. Vizwiz: Nearly real-time answers to visual questions. In *Proceedings of the 23nd Annual ACM Symposium on User Interface Software and Technology*, 2010.

[4] Haotian Liu et al. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[5] Wenliang Dai et al. Instructblip: Towards general-purpose vision-language models with instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[6] Danna Gurari et al. Vizwiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[7] Kristen Grauman et al. Ego4d: Around the world in 3,000 hours of egocentric video. *arXiv preprint arXiv:2110.07058*, 2021.

[8] Abhishek Das et al. Embodied question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[9] Edward J. Hu et al. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2022.

[10] Long Ouyang et al. Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*, 2022.

[11] Rafael Rafailov et al. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023.

[12] Spencer Whitehead et al. Reliable visual question answering: Abstain rather than answer incorrectly. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022.

[13] Lorenz Kuhn et al. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *arXiv preprint arXiv:2302.09664*, 2023.

[14] Tal Schuster et al. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Moonshot AI. Kimi-vl technical report. 2025. `https://github.com/MoonshotAI/Kimi-VL`.

[16] Yaowei Zheng et al. Llama-factory: Unified efficient fine-tuning of 100+ language models. *arXiv preprint arXiv:2403.13372*, 2024.

A Loss Components

Accuracy. Let $S(y, y_k^*) \in [0, 1]$ denote the semantic similarity between the generated response and the k-th reference answer, computed via BERTScore F1 or a GPT-4o-mini judge following EgoBlind's evaluation protocol. The accuracy loss penalizes failure to match any reference answer:

$$\mathcal{L}_{\text{acc}}(y, \mathcal{Y}^*) \;=\; 1 \;-\; \max_{k \in [K]} \; S(y,\, y_k^*). \tag{5}$$

Assistive Utility. EgoBlind provides a utility score $\text{Score}(y, \mathcal{Y}^*) \in [0, 5]$ measuring assistive quality beyond mere lexical correctness. Normalizing to $[0, 1]$:

$$\mathcal{L}_{\text{util}}(y, \mathbf{x}, \mathcal{Y}^*) \;=\; 1 \;-\; \frac{\text{Score}(y,\, \mathcal{Y}^*)}{5}. \tag{6}$$

Latency. Autoregressive generation latency scales approximately linearly with output length; we therefore use token count $|y|$ as a differentiable proxy. The penalty operates in three regimes: zero below a minimum length $\tau_{\min}$ (to discourage degenerate short responses), smooth polynomial growth in the transition zone $(\tau_{\min}, \tau_{\max}]$, and a steep linear overflow penalty beyond the hard cap $\tau_{\max}$:

$$\mathcal{L}_{\text{lat}}(y) \;=\; \begin{cases} 0 & \text{if } |y| \le \tau_{\min} \\ \left(\dfrac{|y| - \tau_{\min}}{\tau_{\max} - \tau_{\min}} \right)^p & \text{if } \tau_{\min} < |y| \le \tau_{\max} \\ 1 + \kappa \left(|y| - \tau_{\max} \right) & \text{if } |y| > \tau_{\max} \end{cases}, \tag{7}$$

where $p \ge 1$ controls the curvature of the transition and $\kappa > 0$ is the overflow slope. When $p = 1$ the transition is linear; $p > 1$ concentrates the penalty near $\tau_{\max}$.

B Tables

Table 1: CLIP urgency classifier performance.

Metric	Validation	Test
Accuracy	0.863	0.798
Precision	0.930	0.879
Recall	0.807	0.695
F1 Score	0.864	0.777
ROC-AUC	0.938	0.905

Table 2: Baseline performance (moonshot-v1-32k-vision-preview, uniform policy).

Metric	Overall	Urgent	Non-urgent
Token-overlap F1	0.1025	0.1245	0.0778
Test examples	4,029	2,142	1,887

Table 3: Per-category composite loss under constrained-optimal grid-search configuration ($\alpha = 0.5, \beta = 0.3, \gamma = 0.2, \tau_{\min} = 8, \tau_{\max} = 40, p = 3, \kappa = 0.3$; test set).

Question Type	Loss	Similarity	N
Safety warnings	1.118	0.118	946
Tool use	0.948	0.139	211
Navigation	0.787	0.117	363
Other resources	0.501	0.112	222
Information reading	0.489	0.089	1,873
Social communication	0.459	0.065	37
Communication & interaction	0.450	0.065	81

Table 4: Baseline vs. fine-tuned (Approach 1, text-only SFT) on the test set.

Metric	**Baseline**	**Fine-tuned**	Δ
Overall loss	3.247	4.570	+1.323
Overall similarity	0.103	0.094	-0.009
Urgent loss	5.644	8.200	+2.556
Non-urgent loss	0.644	0.626	-0.018